

Electromagnetic Waves (Light) — Chapter 30

PHYS 2203 — Introductory Physics III

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Overview

Starting from Maxwell's equations, we will derive the existence of electromagnetic waves and identify visible light as one example. We will see that the speed of light emerges not as an empirical input but as a prediction of electromagnetism. Along the way we will meet the wave equation and its standing- and traveling-wave solutions, and we will close with a survey of the electromagnetic spectrum, from γ -rays to radio waves.

1 Maxwell's Equations

The four equations that govern classical electricity and magnetism are stated together below.

$$\textbf{Gauss's Law:} \quad \oint_S E_n dA = \frac{Q_{\text{inside}}}{\epsilon_0} \quad (1)$$

$$\textbf{Gauss's Law for Magnetism:} \quad \oint_S B_n dA = 0 \quad (2)$$

$$\textbf{Faraday's Law:} \quad \oint_C \vec{E} \cdot d\vec{\ell} = - \int_S \frac{\partial B_n}{\partial t} dA \quad (3)$$

$$\textbf{Ampère's Law:} \quad \oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I + \mu_0 \epsilon_0 \int_S \frac{\partial E_n}{\partial t} dA \quad (4)$$

The symbols:

$\vec{E}$: electric field	$\vec{B}$: magnetic field
E_n : $\vec{E}$ component normal to dA	B_n : $\vec{B}$ component normal to dA
ϵ_0 : permittivity of free space	μ_0 : permeability of free space
Q_{inside} : enclosed charge	I : current through surface
S : surface	C : path

The surface integrals in the last two equations are the **electric flux** and **magnetic flux** through S ,

$$\Phi_E \equiv \int_S E_n dA, \quad \Phi_B \equiv \int_S B_n dA, \quad (5)$$

so for a stationary loop Faraday's and Ampère's laws can be written compactly as

$$\oint_C \vec{E} \cdot d\vec{\ell} = - \frac{d\Phi_B}{dt}, \quad \oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}. \quad (6)$$

These flux forms are the ones we will apply to infinitesimal loops below.

Key observation: Faraday’s law states that a time-varying $\vec{B}$ produces a circulating $\vec{E}$, and Ampère’s law (with Maxwell’s displacement-current term) states that a time-varying $\vec{E}$ produces a circulating $\vec{B}$. The mutual coupling of these two effects is the origin of electromagnetic wave propagation.

2 Parallel Plate Capacitor: Setting Up the Wave

A parallel plate capacitor driven by a sinusoidal voltage $V \sim V_0 \sin \omega t$ provides a region of free space with a time-varying electric field. As the voltage oscillates, so do the plate charges and hence the field: a uniform $\vec{E}_0$ along $\hat{y}$ between the plates, and a fringing field $\vec{E}_{\text{fringe}}$ bowing outward at the edges. In the surrounding vacuum, $\partial \vec{E} / \partial t \neq 0$.

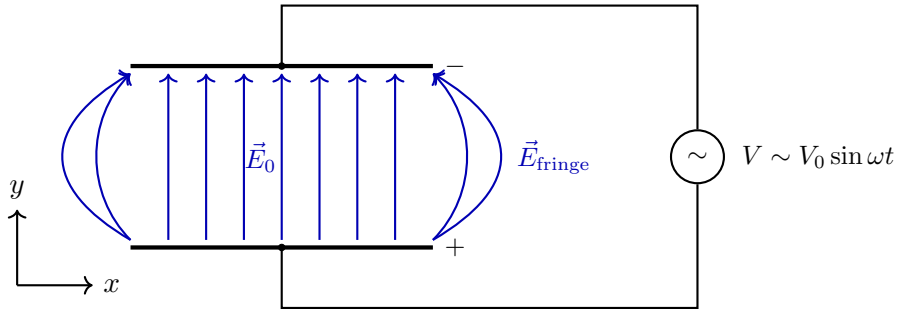


Figure 1: Parallel plate capacitor driven by a sinusoidal voltage source. Uniform $\vec{E}_0$ between the plates, fringing fields $\vec{E}_{\text{fringe}}$ at the edges.

Our goal: find the $\vec{B}$ field that accompanies this time-varying $\vec{E}$, and show that the two propagate together as a wave.

Plane-wave simplification

Away from the capacitor we track a disturbance propagating along $+\hat{x}$ and assume the simplest field configuration consistent with it: each field has a single component, depending only on x and t ,

$$\vec{E} = E_y(x, t) \hat{y}, \quad \vec{B} = B_z(x, t) \hat{z}. \quad (7)$$

We now apply Faraday’s and Ampère’s laws to two infinitesimal loops at a representative point — one in the x – y plane, one in the x – z plane. Each yields one PDE, and the two PDEs combine into a wave equation.

3 Applying Faraday’s Law (Loop in the x – y Plane)

Apply Faraday’s law, $\oint \vec{E} \cdot d\vec{\ell} = -d\Phi_B/dt$, to an infinitesimal rectangle in the x – y plane at $(x, 0, 0)$: width dx along $\hat{x}$, height ℓ along $\hat{y}$, traversed counter-clockwise starting at the top-left corner ($\bullet$).

Fields on this loop:

- On the *left* edge (at x) the field is $E_y(x)$; on the *right* edge (at $x + dx$) it has grown to $E_y(x + dx) = E_y + dE_y$.

- The magnetic field B_z threads the loop, out of the page.

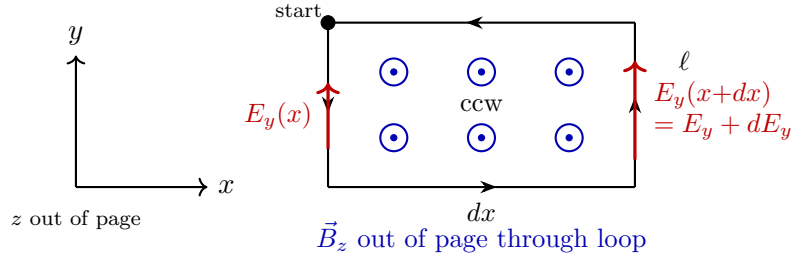


Figure 2: Infinitesimal Faraday loop in the x - y plane at $(x, 0, 0)$. The red arrows show the field E_y on the two vertical edges (slightly larger on the right since E_y varies with x). The blue $\odot$ symbols show B_z pointing out of the page through the loop. The path is traversed counter-clockwise starting at the top-left corner: $-\hat{y}$ (left edge), $+\hat{x}$ (bottom), $+\hat{y}$ (right edge), $-\hat{x}$ (top).

Walk around the loop segment by segment, keeping track of the direction of travel on each edge. Since $\vec{E}$ has only a y -component, only the two vertical edges contribute:

1. **Top-left $\rightarrow$ bottom-left** (left edge, $-\hat{y}$, length ℓ).
 $\vec{E} = E_y(x) \hat{y}$, $d\vec{\ell} = -\hat{y} d\ell$. **Term:** $-E_y \ell$.
2. **Bottom-left $\rightarrow$ bottom-right** ($+\hat{x}$, length dx).
 $E_x = 0$. **Term:** 0.
3. **Bottom-right $\rightarrow$ top-right** (right edge, $+\hat{y}$, length ℓ).
 $\vec{E} = (E_y + dE_y) \hat{y}$, $d\vec{\ell} = +\hat{y} d\ell$. **Term:** $+(E_y + dE_y) \ell$.
4. **Top-right $\rightarrow$ top-left** ($-\hat{x}$, length dx).
 $E_x = 0$. **Term:** 0.

The $\pm E_y \ell$ pieces cancel:

$$\oint \vec{E} \cdot d\vec{\ell} = -E_y \ell + (E_y + dE_y) \ell = dE_y \ell. \quad (8)$$

Right-hand side — the flux. A counter-clockwise path has normal $+\hat{z}$ (right-hand rule), so $\Phi_B = B_z \ell dx$, and Faraday's law reads

$$dE_y \ell = -\frac{d}{dt}(B_z \ell dx). \quad (9)$$

Promoting to partial derivatives. At fixed time, $dE_y = (\partial E_y / \partial x) dx$; the loop is stationary, so the flux changes only through $\partial B_z / \partial t$. Dividing by ℓdx :

$$\boxed{\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t}} \quad (10)$$

Physical interpretation: a spatially varying E_y must be accompanied by a time-varying B_z . This is the first of the two equations we need.

4 Applying Ampère's Law (Loop in the x - z Plane)

Faraday gave one PDE relating E_y and B_z ; Ampère's law gives the second. No charge flows through the vacuum, so $I = 0$ and only the displacement-current term contributes.

Apply Ampère's law to an infinitesimal rectangle in the x - z plane at $(x, 0, 0)$: width dx along $\hat{x}$, height ℓ along $\hat{z}$. **Orientation matters here:** with x to the right and z up in the figure, right-handed axes put $+\hat{y}$ *into* the page, so E_y threads the loop as $\otimes$. We choose the loop normal along $+\hat{y}$ (parallel to $\vec{E}$); the right-hand rule then fixes the traversal shown, starting at the bottom-left corner ($\bullet$) and going up the left edge first.

Fields on this loop:

- On the *left* edge (at x) the magnetic field is $B_z(x)$; on the *right* edge (at $x + dx$) it is $B_z(x + dx) = B_z + dB_z$.
- The electric field E_y threads the loop, into the page ($\otimes$).

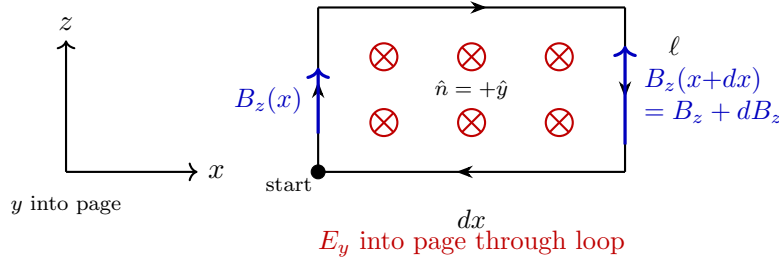


Figure 3: Infinitesimal Ampère loop in the x - z plane at $(x, 0, 0)$. The blue arrows show B_z on the two vertical edges (larger on the right since B_z varies with x). The red $\otimes$ symbols show E_y threading the loop along $+\hat{y}$, which points *into* the page for right-handed axes drawn with x right and z up. The traversal (start at bottom-left: $+\hat{z}$ up the left edge, $+\hat{x}$ along the top, $-\hat{z}$ down the right edge, $-\hat{x}$ along the bottom) is right-handed about the normal $\hat{n} = +\hat{y}$.

Walk around the loop segment by segment. Since $\vec{B}$ has only a z -component, only the two vertical edges contribute:

1. **Bottom-left $\rightarrow$ top-left** (left edge, $+\hat{z}$, length ℓ).
 $\vec{B} = B_z(x) \hat{z}$, $d\vec{\ell} = +\hat{z} d\ell$. **Term:** $+B_z \ell$.
2. **Top-left $\rightarrow$ top-right** ($+\hat{x}$, length dx).
 $B_x = 0$. **Term:** 0.
3. **Top-right $\rightarrow$ bottom-right** (right edge, $-\hat{z}$, length ℓ).
 $\vec{B} = (B_z + dB_z) \hat{z}$, $d\vec{\ell} = -\hat{z} d\ell$. **Term:** $-(B_z + dB_z) \ell$.
4. **Bottom-right $\rightarrow$ bottom-left** ($-\hat{x}$, length dx).
 $B_x = 0$. **Term:** 0.

The $\pm B_z \ell$ pieces cancel:

$$\oint \vec{B} \cdot d\vec{\ell} = B_z \ell - (B_z + dB_z) \ell = -dB_z \ell. \quad (11)$$

Right-hand side — the displacement current. With $I = 0$ and normal $+\hat{y}$, the electric flux is $\Phi_E = E_y \ell dx$, and Ampère’s law reads

$$-dB_z \ell = \mu_0 \varepsilon_0 \frac{d}{dt}(E_y \ell dx). \quad (12)$$

Promoting to partial derivatives as before ($dB_z = (\partial B_z / \partial x) dx$ at fixed time) and dividing by ℓdx :

$$\boxed{-\frac{\partial B_z}{\partial x} = \mu_0 \varepsilon_0 \frac{\partial E_y}{\partial t}} \quad (13)$$

Physical interpretation: a spatially varying B_z must be accompanied by a time-varying E_y — the counterpart of the Faraday result. We now have two equations relating E_y and B_z , so we can eliminate one field and solve for the other.

5 Deriving the Wave Equation

We have two coupled first-order PDEs:

$$\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t} \quad \text{and} \quad -\frac{\partial B_z}{\partial x} = \mu_0 \varepsilon_0 \frac{\partial E_y}{\partial t}.$$

Strategy: differentiate each once more — with respect to x and with respect to t — then eliminate one field by substitution, leaving a single second-order PDE for each field alone.

$$\textcircled{1} \quad \frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t} \implies \frac{\partial^2 E_y}{\partial x^2} = -\frac{\partial^2 B_z}{\partial x \partial t} \text{ and } \frac{\partial^2 E_y}{\partial x \partial t} = -\frac{\partial^2 B_z}{\partial t^2} \textcircled{2} \quad (14)$$

$$\textcircled{3} \quad -\frac{\partial B_z}{\partial x} = \mu_0 \varepsilon_0 \frac{\partial E_y}{\partial t} \implies -\frac{\partial^2 B_z}{\partial x \partial t} = \mu_0 \varepsilon_0 \frac{\partial^2 E_y}{\partial t^2} \text{ and } -\frac{\partial^2 B_z}{\partial x^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_y}{\partial t \partial x} \textcircled{4} \quad (15)$$

Combine ① and ③ (the mixed partials $\partial^2 B_z / \partial x \partial t$ are equal):

$$\boxed{\frac{\partial^2 E_y}{\partial t^2} = \frac{1}{\mu_0 \varepsilon_0} \frac{\partial^2 E_y}{\partial x^2}} \quad (16)$$

Combine ② and ④:

$$\boxed{\frac{\partial^2 B_z}{\partial t^2} = \frac{1}{\mu_0 \varepsilon_0} \frac{\partial^2 B_z}{\partial x^2}} \quad (17)$$

These are **wave equations** with speed

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 2.998 \times 10^8 \text{ m/s} \quad \text{“speed of light”}} \quad (18)$$

The constants ε_0 and μ_0 are measured in tabletop experiments — ε_0 from Coulomb’s law, μ_0 from forces between current-carrying wires — with no light involved anywhere. Yet $1/\sqrt{\mu_0 \varepsilon_0}$ reproduces the measured speed of light. This result, first obtained by Maxwell in 1865, identifies light as an electromagnetic wave.

6 Wave Equation in Other Physical Systems

The same equation,

$$\frac{\partial^2 f(x, t)}{\partial t^2} = v^2 \frac{\partial^2 f(x, t)}{\partial x^2}, \quad (19)$$

describes many physical systems:

Example 1 — Guitar string: $f(x, t)$ = transverse displacement of stretched string,

$$v = \sqrt{\frac{\text{string tension}}{\text{mass/length of string}}} \quad (20)$$

Example 2 — Sound waves in air: $f(x, t)$ = air pressure,

$$v = \text{speed of sound in air } (\approx 340 \text{ m/s}) \quad (21)$$

Because the equation is identical, so are its solutions: whatever we learn about $f(x, t)$ applies directly to $E_y(x, t)$ and $B_z(x, t)$.

7 Solutions to the Wave Equation

What functions solve the wave equation? Since it is second order in both x and t , we expect a four-parameter family of sinusoidal solutions. Two physically distinct forms are useful: the **standing wave** (fixed spatial profile, amplitude oscillating in time) and the **traveling wave** (profile moving rigidly at speed v). We verify each by direct substitution.

Standing Wave Solution

$$\boxed{f(x, t) = A \sin(kx + \theta) \cos(\omega t + \phi)} \quad \text{where } v = \frac{\omega}{k} \quad (22)$$

The four constants: A (amplitude), k (wave number), θ and ϕ (phases); ω is fixed by $\omega = vk$.

Verify by direct substitution:

$$\frac{\partial^2 f(x, t)}{\partial t^2} = -\omega^2 f(x, t) \quad (23)$$

$$\frac{\partial^2 f(x, t)}{\partial x^2} = -k^2 f(x, t) \quad (24)$$

Plugging into the wave equation:

$$-\omega^2 f(x, t) = v^2 (-k^2 f(x, t)) \implies v^2 = \frac{\omega^2}{k^2} \quad \checkmark \quad (25)$$

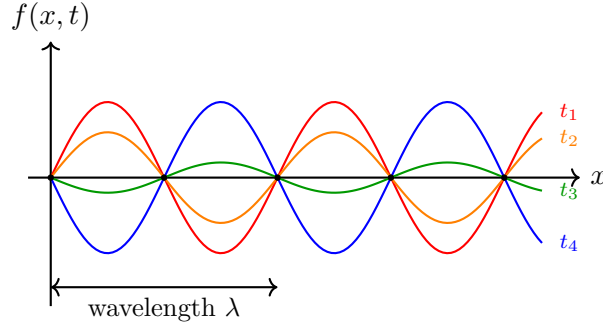


Figure 4: Standing wave snapshots at four times $t_1 < t_2 < t_3 < t_4$. Nodes (dots) are stationary; the envelope oscillates. The period of the example is $T = 2(t_4 - t_1)$.

Key quantities:

$$\text{wave number: } k = \frac{2\pi}{\lambda} \quad [\text{m}^{-1}] \quad (26)$$

$$\text{angular frequency: } \omega = \frac{2\pi}{T} \quad [\text{s}^{-1}] \quad (27)$$

$$\text{frequency: } f = \frac{1}{T} \quad [\text{Hz}] \quad (28)$$

Once the medium fixes v , wavelength and period are not independent: $v = \omega/k = \lambda/T = f\lambda$.

Traveling Wave Solution

A traveling wave depends on x and t only through the combination $kx - \omega t$: a point of constant phase moves at $dx/dt = \omega/k = v$, so the whole profile slides rigidly at speed v .

$$\boxed{f(x, t) = A \sin(kx - \omega t + \delta)} \quad \text{where } v = \frac{\omega}{k} \quad (29)$$

Verify by direct substitution:

$$\frac{\partial^2 f(x, t)}{\partial t^2} = -\omega^2 f(x, t) \quad (30)$$

$$\frac{\partial^2 f(x, t)}{\partial x^2} = -k^2 f(x, t) \quad (31)$$

Plugging into the wave equation gives the same condition $v = \omega/k \Rightarrow$ solution verified.

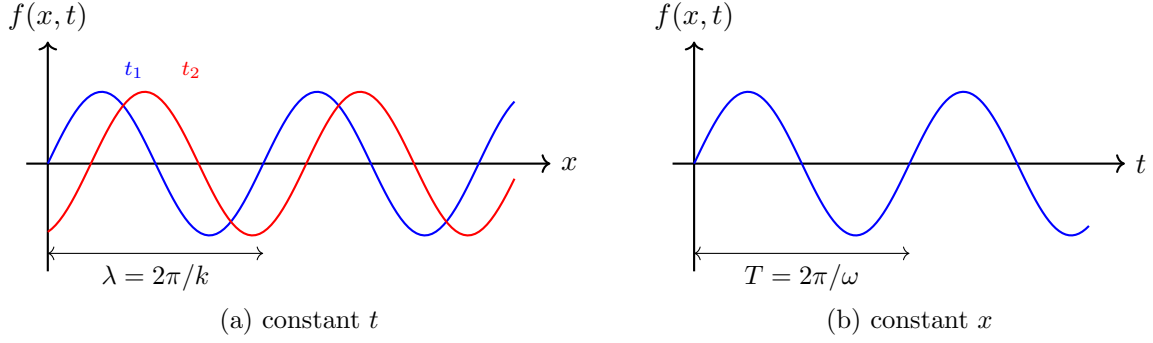


Figure 5: (a) Two snapshots of a traveling wave at t_1 and $t_2 > t_1$; the waveform shifts to the right by $v(t_2 - t_1)$. (b) The same wave observed at constant x vs. t , with period $T = 2\pi/\omega$.

Conventions:

- Usually choose $t = 0$ such that $\delta = 0$.
- $f(x, t) = A \sin(kx - \omega t)$ describes a wave traveling in the $+x$ direction with speed $v = \omega/k$, where A is the amplitude.
- $f(x, t) = A \sin(kx + \omega t)$ describes a wave traveling in the $-x$ direction with speed $v = \omega/k$.

8 Application to EM Waves

Each wave equation can be solved on its own, but E_y and B_z are not independent — they must also satisfy the original first-order equations (Faraday and Ampère). As we now show, this fixes the relative phase and the ratio of the amplitudes.

The wave equations for E_y and B_z in vacuum are:

$$\frac{\partial^2 E_y}{\partial t^2} = c^2 \frac{\partial^2 E_y}{\partial x^2} \quad \text{and} \quad \frac{\partial^2 B_z}{\partial t^2} = c^2 \frac{\partial^2 B_z}{\partial x^2} \quad (32)$$

Take traveling wave solutions:

$$E_y(x, t) = E_0 \sin(kx - \omega t + \delta), \quad B_z(x, t) = B_0 \sin(kx - \omega t + \tilde{\delta}) \quad (33)$$

Recall Faraday's Law requires $\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t}$:

$$kE_0 \cos(kx - \omega t + \delta) = +\omega B_0 \cos(kx - \omega t + \tilde{\delta}) \quad \text{for all } x \text{ and } t. \quad (34)$$

This forces $\delta = \tilde{\delta}$ (usually set $\delta = \tilde{\delta} = 0$) **and**

$$E_0 = \frac{\omega}{k} B_0 \implies \boxed{E_0 = cB_0} \quad (35)$$

Therefore:

$$\boxed{E_y(x, t) = E_0 \sin(kx - \omega t)} \quad \boxed{B_z(x, t) = B_0 \sin(kx - \omega t)} \quad (36)$$

with $c = \frac{\omega}{k} = f\lambda$.

Three principal features of this result:

1. **Common phase.** $\vec{E}$ and $\vec{B}$ reach their maxima together, vanish together, and reverse together; they are in phase.
2. **Perpendicular geometry.** $\vec{E}$ lies along $\hat{y}$, $\vec{B}$ lies along $\hat{z}$, and propagation is along $\hat{x}$. The three directions are mutually perpendicular, identifying the wave as transverse. Electromagnetic waves in vacuum are always transverse.
3. **Fixed amplitude ratio.** $E_0 = cB_0$. For a charge q moving at speed v , the magnetic force is at most $qvB_0 = (v/c)qE_0$ — smaller than the electric force by a factor v/c . Since $v \ll c$ for ordinary charges, the force exerted by a light wave on a charged particle is dominated by the electric field.

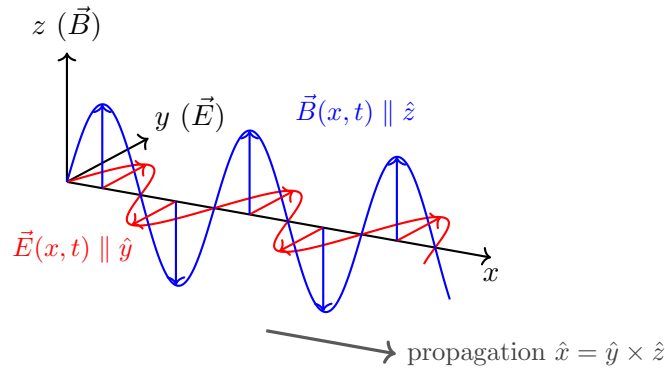


Figure 6: Plane EM wave traveling in $+\hat{x}$: $\vec{E}$ oscillates along $\hat{y}$, $\vec{B}$ along $\hat{z}$, in phase, with $E_0 = cB_0$. The triad $(\hat{x}, \hat{y}, \hat{z})$ is right-handed and the direction of propagation equals $\hat{E} \times \hat{B} = \hat{y} \times \hat{z} = \hat{x}$.

9 Electromagnetic Spectrum

The wave equation contains no preferred wavelength: every λ (equivalently every frequency $f = c/\lambda$) is an allowed solution. The named bands differ only in how the radiation is produced and detected — the underlying physics is identical. The full spectrum, from short to long wavelength:

γ -rays	$\lesssim 10^{-11}$ m
x-rays	$\sim 10^{-11}$ – 10^{-8} m
ultraviolet	~ 10 – 400 nm
visible	400 – 700 nm
infrared	700 nm– 1 mm
microwave	~ 1 mm– 1 m
TV, FM	~ 1 – 10 m
AM radio	$\sim 10^2$ – 10^3 m

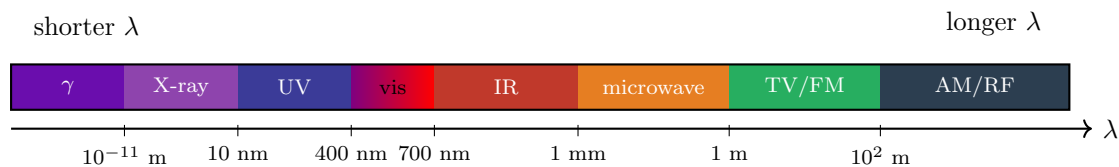


Figure 7: Electromagnetic spectrum spanning γ -rays through AM/RF, with approximate boundary wavelengths marked on the axis. There are no restrictions on the value of λ .

The visible band occupies only a narrow portion of the spectrum, located near the peak of the Sun's emission and within a transparent window of the Earth's atmosphere. The other bands are just as physical: radio for communication, microwaves for heating and radar, infrared for thermal imaging, X-rays for medical and structural imaging. All propagate in vacuum at the same speed c .